

15 AI in the research process

As the agents became more capable, my work shifted from directing individual steps toward specifying what the finished work had to accomplish. They could develop a substantial mathematical argument from an incomplete suggestion, yet sometimes needed repeated intervention to pursue a much more ordinary objective. My main difficulty increasingly concerned what they treated as success. This section gives my interpretation of that experience; the separate disclosure records the extent and provenance of AI use in the thesis.

15.1 A mathematical objective with room for discovery

The local maximality proof illustrates how much could be delegated without specifying a procedure. I asked a GPT-5-series model to develop a local maximality argument for HKO in facet coordinates. I suggested looking for several upper bounds and somehow showing that together they control a neighborhood after quotienting out the symmetries. I had neither specified a finite certificate nor worked out how to construct the upper bounds.

The model developed essentially the entire argument from that starting point. It identified a sufficient configuration of twenty-six upper bounds in the twenty-five transverse dimensions. An initial derivative-based approach had provided only twenty-five nonsingular cases. The model then used the weights β in the quadratic objective to construct feasible sections: an invertible 5×5 closure minor lets five weights depend smoothly on the facet parameters while the remaining weights stay fixed. This supplied upper bounds even where the initial approach failed. It also completed proof steps involving the volume and wrote the SageMath code for the exact computations.

The desired result was clear enough to guide this work even though the route to it was unknown. A proof had to establish the local inequality, including directions that did not preserve the product structure. The model was free to discover what finite statement would suffice. In particular, the twenty-six derivative conditions were part of its solution, not a test suite that I supplied in advance. The resulting ten-facet theorem in this thesis shows the value of giving an agent a mathematical objective while leaving substantial freedom over the argument.

15.2 An inferred criterion of success

Other tasks produced a more puzzling contrast. An agent could reason competently about a problem and still stop after a first step, answer a related question, or keep preparing work that it had been asked to finish. These failures sometimes looked oddly motivational: the next useful action seemed to be within the agent’s capabilities, but its behavior was directed toward a different stopping point. I found it useful to think about the criterion of success that the agent appeared to infer from the conversation.

My hypothesis is that training with verifiable rewards encourages models to learn a concept of a *Grader*: a representation of what will judge the task completed successfully. Reinforcement learning with verifiable rewards uses outcomes that can be checked, such as a mathematical answer or a program’s performance on test cases. For example, DeepSeek-R1-Zero was trained with rule-based accuracy and format rewards; its reasoning strategies were not supplied as complete human demonstrations [GYZ+25]. I suspect that such training can teach a model to infer a criterion and organize its work around satisfying it. The training result motivates this interpretation, but does not establish that it explains the behavior of the agents I used.

This interpretation concerns more than reward hacking or literal obedience to an instruction. An agent need not exploit a loophole in a stated test, or misread a particular word, to pursue the wrong objective. A plan for improving a chapter can be a good answer to a conversational turn while leaving the requested chapter unwritten. Likewise, a useful preliminary calculation can become the apparent endpoint of a task that required a theorem. Each answer may be

sensible on its own. The failure is that the criterion organizing the work covers only part of the intended result.

A clear objective makes the inferred criterion more likely to coincide with what I actually want. Here clarity means enough specificity to evaluate completion: what result is needed, why it is needed, which constraints matter, and what would leave the task unfinished. It does not require prescribing every intermediate action. The HKO example was effective precisely because the mathematical objective constrained the result while allowing the agent to invent the method. A prompt does not update the model’s parameters; it supplies the context in which these learned capabilities are used.

15.3 Keeping the whole task in view

Long conversations made the distinction between a useful answer and a completed task increasingly important. Some of my threads extended beyond a hundred turns. Habits inherited from short, fast exchanges became a poor fit: each new message could receive a plausible response while earlier constraints, unresolved work, and the reason for the project receded from view. Missing context or a confused description of the goal could leave an otherwise capable agent passive, or busy with work that no longer served the main objective.

The checkpoint I want in such a task asks whether the current reasoning, evidence, and actions are advancing the right whole goal within the available budget. It should expose a change of destination: delivering only a first step, solving an easier related problem, or replacing completion by repeated planning. The same question belongs at the end. A successful evaluation of several components does not answer whether the requested result has been delivered.

The budget is part of that judgment. An investigation can produce useful information and still be a poor choice if it consumes the time needed to finish the work. When the task is to deliver a readable chapter, identifying another possible improvement has to be weighed against writing and revising the actual chapter. When the task is a mathematical search, the value of another computation depends on the question it can resolve and the resources left for following up. Progress has to be assessed against the intended outcome and its constraints, rather than the amount of activity generated.

This does not call for inspecting every step. An autonomous agent should choose and revise many of its own intermediate steps. The important human intervention is often to restore the purpose that those steps serve. More detailed instructions can help when a constraint was missing, but can also obscure the goal beneath local requirements. I increasingly needed to explain the result I wanted and the context that made it useful, then judge whether the work still addressed that result.

15.4 Why exposition remained difficult

For the mathematics required by this MSc project, I often found the agents’ capabilities sufficient to make substantial progress with little detailed direction. Producing passable mathematical exposition was much harder to delegate. This was surprising to me given the amount of written material used in language-model training. The later work on the thesis became a prolonged struggle with writing and task-management quirks, even when the models could partly recognize those quirks in discussion.

One recurring problem was prose that presented itself as polished before it communicated its content clearly. Inflated claims of significance, colorful phrasing, and abstract summaries could make a passage sound complete while leaving its mathematical point difficult to recover. This is what I mean by *AI slop* in the present context. Scientific exposition instead needs to show what is being claimed, what makes it true, and why that claim belongs at this point in the argument. An agent’s ability to decipher a dense paragraph did not show that a reader could comfortably follow it.

I suspect that preference training contributes to this tendency. In reinforcement learning from human feedback, rankings of model outputs can be used to train the model toward preferred responses, as in InstructGPT [OWJ+22]. If an impressive presentation is easier to recognize in a brief comparison than a clear explanation is to assess through sustained reading, that preference signal may favor the appearance of quality. This is my hypothesis about a possible source of the prose I encountered, not a demonstrated causal account of those models’ training.

The practical difficulty persisted even after it had been named. A model could acknowledge that a passage needed a concrete argument and then return another description of the argument’s intended role. It could recognize that the goal was a readable chapter while spending its effort on further diagnosis. I had to devise corrections tailored to recurring behaviors; there was no single instruction whose recognition reliably produced the desired writing. Understanding a criticism in one exchange and carrying it through a sustained piece of work proved to be different capabilities.

This experience also limits the lesson I draw from the mathematical success. The proof work in this project was tractable enough for the available models that their quirks often mattered less there. The difficulty of exposition was not captured by the formal sophistication of its subject. Choosing the right explanation, maintaining its purpose across a chapter, and removing language that obscured the mathematics remained demanding tasks.

15.5 The changing human contribution

The human work therefore shifted toward translating the needs of the research project into objectives an agent could pursue and evaluate. I knew why I wanted a theorem, an experiment, or a thesis chapter; that purpose did not automatically accompany a request for one. Supplying it meant explaining the intended result, the reader or use it had to serve, and the time and resources available. It also meant compensating for behaviors that remained unreliable despite the models’ growing autonomy.

This left less need to manage every mathematical or programming step and more need to keep the intended outcome clear. Choosing worthwhile questions and judging what the answers meant remained part of the shared work. In this project, substantial autonomous problem solving and substantial human intervention coexisted. The challenge was increasingly to make the task the agent pursued coincide with the task I needed completed.

References

- [GYZ+25] Daya Guo, Dejian Yang, Haowei Zhang, et al. “DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning”. In: *Nature* 645 (2025), pp. 633–638. DOI: 10.1038/s41586-025-09422-z.
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